Governance of Explainable AI in Internal Financial Control Systems: A Proposed Framework Integrating XAI Techniques with the COSO Model to Enhance Institutional Sustainability

Ali Abdulwahab Yahya Al-Saffar[a,*]

[a] *College of Administration and Economics, University of Mosul, Mosul, Iraq*

* Corresponding author. E-mail: ali_alsaffar@uomosul.edu.iq. ORCID: 0009-0005-0918-893X

Abstract

Machine-learning models now sit inside many internal financial controls, yet their opacity is hard to square with the transparency and accountability that control frameworks demand. This paper proposes IFC-XAI, a governance framework that ties the outputs of explainable artificial intelligence (XAI) to the five components of the COSO Internal Control–Integrated Framework (2013), and asks whether the idea holds up on real data. Following a Design Science Research approach, four classifiers (logistic regression, decision tree, random forest and gradient boosting) were trained on the public German Credit dataset (n = 1000; 20 raw and 61 encoded features). On a stratified hold-out set (n = 300) logistic regression gave the highest area under the ROC curve (AUC = 0.801), with random forest next (0.797). Under repeated stratified cross-validation (5 folds × 3 repeats) the two could not be separated statistically (paired t-test, p = 0.058). The SHAP attributions of the logistic regression model were then summed by COSO component. Risk Assessment carried 55.9% of the explanatory weight among the top 15 features, Control Activities 28.2% and Control Environment 15.9%. Moving the decision threshold from 0.50 to 0.30 raised recall from 0.533 to 0.733 and cut missed risky cases from 42 to 24, although false alarms rose from 23 to 59. Where to set the threshold is therefore a decision that a governance process should record. The paper offers a COSO-aligned architecture for explainability, a reproducible mapping protocol, and evidence from a public benchmark. The data are old and of foreign origin, the sample is small, and the mapping of features to COSO components was defined by the author and has not been validated.

Keywords: explainable artificial intelligence; COSO framework; internal financial control; SHAP; AI governance; institutional sustainability

Highlights

- A four-layer framework connects explainable AI outputs to the five COSO components.
- SHAP attributions from credit data are summed by COSO component.
- An interpretable logistic regression performs on a par with ensembles under repeated cross-validation.
- The choice of decision threshold is treated as a governance decision that must be documented.
- Code and data are open, so every result can be reproduced.

1. Introduction

Artificial intelligence (AI) has moved quickly from pilot projects to everyday use in internal financial control, where it supports fraud detection, credit-risk assessment and continuous auditing (Kokina & Davenport, 2017; Sun & Vasarhelyi, 2018). The models that perform best are often the hardest to read. An auditor or audit committee who wants to know why an automated control flagged one transaction and passed another may have no way to find out. That is uncomfortable for a profession built on transparency, accountability and professional judgment (Munoko et al., 2020; Zhang et al., 2022).

The COSO Internal Control–Integrated Framework (COSO, 2013) is the most widely used reference for designing and assessing internal control. It has five components: control environment, risk assessment, control activities, information and communication, and monitoring activities. The framework was written with human-operated controls in mind. Once a learning algorithm performs part of a control, someone has to decide how its decisions will be explained and who answers for them. Professional bodies have started to publish guidance on applying enterprise risk-management and risk-governance frameworks to AI (COSO & Deloitte, 2021; NIST, 2023), but the explanation techniques themselves still sit largely outside those frameworks.

Explainable AI (XAI) tackles the opacity problem with tools such as Shapley-value attributions (Lundberg & Lee, 2017) and local surrogate models (Ribeiro et al., 2016), and the literature on them is large and well surveyed (Guidotti et al., 2018; Arrieta et al., 2020). Little work was found, however, that takes quantitative XAI output, restates it in the vocabulary of a standard internal-control framework, and checks the result against data. This paper tries to fill that gap.

It makes three contributions. The first is IFC-XAI, a four-layer governance framework whose third layer spreads explanation outputs across the five COSO components. The second is a reproducible protocol for summing SHAP attributions by COSO component. The third is empirical evidence from a public benchmark, accompanied by robustness checks (repeated cross-validation, bootstrap confidence intervals and a decision-threshold analysis) that show how far the results can be pushed.

Section 2 covers the background and related work, and Section 3 the methodology. The framework is presented in Section 4 and the results in Section 5. Section 6 discusses what follows from them, Section 7 lists limitations and future work, and Section 8 concludes.

2. Background and related work

2.1 The COSO internal control framework

COSO (2013) describes internal control as a process carried out by the board, management and other personnel, and meant to give reasonable assurance that objectives for operations, reporting and compliance will be met. Table 1 lists the five components.

Table 1. The five components of the COSO Internal Control–Integrated Framework (COSO, 2013).

Component	Brief description
Control environment	Integrity, ethical values and the organizational structure that underpin control
Risk assessment	Identification and analysis of risks to the achievement of objectives
Control activities	Policies and procedures that ensure management directives are carried out
Information and communication	Timely exchange of relevant information internally and externally
Monitoring activities	Ongoing or periodic evaluation of whether the components are present and functioning

2.2 Explainable artificial intelligence

XAI covers the methods that make the decisions of complex models understandable to people (Doshi-Velez & Kim, 2017). Surveys sort these methods by scope (global or local) and by whether they depend on the model (model-specific or model-agnostic) (Adadi & Berrada, 2018; Arrieta et al., 2020). Two techniques dominate. SHAP rests on the Shapley value from cooperative game theory (Shapley, 1953). It

divides the gap between a prediction and the average prediction among the input features in proportion to their marginal contributions (Lundberg & Lee, 2017), and fast variants exist for tree ensembles (Lundberg et al., 2020). LIME instead fits a simple linear model around a single prediction to approximate how the complex model behaves there (Ribeiro et al., 2016).

The field has its critics. Lipton (2018) pointed out that "interpretability" means different things to different authors, so any claim about it needs a clear statement of what is being explained and to whom. Rudin (2019) went further and argued that post-hoc explanations of black-box models can misrepresent what the model actually does, and that high-stakes decisions should rely on models that are interpretable by design. Both points are taken seriously here. The reference model is an inherently interpretable one, and SHAP is used to aggregate and map attributions onto control components, not to vouch for an opaque model.

Explanation also matters for legal accountability. Goodman and Flaxman (2017) discussed what transparency obligations automated decision-making might create under European regulation, and the NIST AI Risk Management Framework counts explainability and interpretability among the traits of trustworthy AI systems (NIST, 2023).

2.3 Related work in auditing and credit risk

In auditing, Kokina and Davenport (2017) looked at how automation and AI are changing the work, while Sun and Vasarhelyi (2018) showed how deep learning can be used to analyze text in audit settings. Munoko et al. (2020) examined the ethical side of AI in auditing and observed that a lack of transparency creates trouble for accountability and for the auditor's professional judgment. Zhang et al. (2022) put forward a framework for using XAI in auditing and listed open research needs. On the credit side, Bussmann et al. (2021) used a Shapley-value approach to explain credit-scoring models, which shows that such explanations are feasible in financial settings.

At the level of governance, COSO and Deloitte (2021) issued guidance on applying the COSO enterprise risk management framework to AI, and NIST (2023) released a risk-management framework for AI systems. Taken together, these works deal with audit practice, with governance in general, and with the technical explanation of models, but each does so separately. In the literature reviewed informally, the quantitative link between XAI outputs and the five components of the COSO internal-control framework had received little empirical attention. That judgment comes from a non-systematic review and makes no claim to be exhaustive.

3. Methodology

The study follows the Design Science Research (DSR) paradigm (Hevner et al., 2004), in which an artifact is built and then evaluated. The work proceeded in five steps: identifying the problem, designing the conceptual framework, instantiating it on real data, evaluating predictive performance and explanations, and finally discussing the findings and drawing recommendations.

3.1 Data

The German Credit (Statlog) dataset comes from the UCI Machine Learning Repository (Hofmann, 1994; Dua & Graff, 2019). It describes 1000 loan applicants through 20 attributes (for example checking-account status, credit history, loan purpose, savings, employment duration, housing, property and age) and a binary credit outcome. The label was recoded so that 1 means credit risk (default; n = 300, 30.0%) and 0 means no risk (n = 700, 70.0%). Amounts are in Deutsche Marks (DM) because of where the data came from. Table 2 gives descriptive statistics for the numeric attributes. No values are missing.

Table 2. Descriptive statistics of the numeric attributes (n = 1000).

Variable	Mean	SD	Min	Median	Max
Duration (months)	20.90	12.06	4.0	18.0	72.0
Credit amount (DM)	3271.26	2822.74	250.0	2319.5	18424.0
Installment rate	2.97	1.12	1.0	3.0	4.0
Present residence (years)	2.85	1.10	1.0	3.0	4.0
Age (years)	35.55	11.38	19.0	33.0	75.0
Existing credits	1.41	0.58	1.0	1.0	4.0
People liable	1.16	0.36	1.0	1.0	2.0

3.2 Preprocessing and experimental design

Categorical attributes were one-hot encoded and numeric attributes standardized, which produced 61 features. The data were split into a training set (n = 700) and a test set (n = 300) by stratified sampling with a fixed random seed (42). To prevent information leakage, preprocessing was fitted inside each training partition only. The analyses ran in Python with scikit-learn 1.9.1 (Pedregosa et al., 2011), SHAP 0.53.0, NumPy 2.5.3, pandas 3.0.5 and SciPy 1.18.1.

3.3 Models and evaluation

Four classifiers were trained. Logistic regression (Hosmer et al., 2013) served as the interpretable reference. The others were a decision tree (maximum depth 6), a random forest (Breiman, 2001; 300 trees, maximum depth 8) and gradient boosting (Friedman, 2001; 200 estimators, maximum depth 3). Hyperparameters were fixed in advance and not tuned, so the comparison shows how reasonable default settings perform and not what each model could reach when optimized.

On the hold-out set, performance was measured by accuracy, precision, recall, F1-score, the area under the ROC curve (AUC; Fawcett, 2006) and the confusion matrix. Robustness was checked through repeated stratified 5-fold cross-validation with three repeats (Kohavi, 1995), a paired t-test on fold-level AUC, a 95% bootstrap confidence interval for the test AUC of logistic regression (2000 resamples; Efron & Tibshirani, 1993), and an analysis of decision thresholds from 0.20 to 0.50.

3.4 Explanation and mapping to COSO

SHAP attributions for the logistic regression model were computed with the linear explainer. When a model is linear and the features are treated as independent, the attribution of feature i for instance x is $\varphi_i(x) = \beta_i(x_i - E[x_i])$. A feature's global importance was taken as its mean absolute attribution over the test set. The 15 most important encoded features were then assigned to COSO components according to a functional classification defined by the author (Table 7), and the explanatory weight of component c was computed as $\text{Share}(c) = \Sigma_{j \in c}\, \text{mean}|\varphi_j| \,/\, \Sigma_{j \in \text{top15}}\, \text{mean}|\varphi_j|$.

This classification is a heuristic. Domain experts have not validated it, and its purpose is to demonstrate the protocol, not to propose a definitive taxonomy. The rule was simple. Indicators of the applicant's financial standing and credit history went to Risk Assessment. Parameters of the loan structure that a lending policy can limit (amount, duration, installment rate, number of existing credits) went to Control Activities. Attributes describing the applicant personally (personal status, employment duration, housing) went to Control Environment, since they raise questions of policy, ethics and fairness that belong under the integrity and ethical-values principles of that component.

4. The proposed IFC-XAI framework

The framework starts from one idea: explainability should be designed in as the link between an intelligent model and the governance system from the start, not added once the model is running. It has four successive layers joined by a continuous feedback loop (Fig. 1).

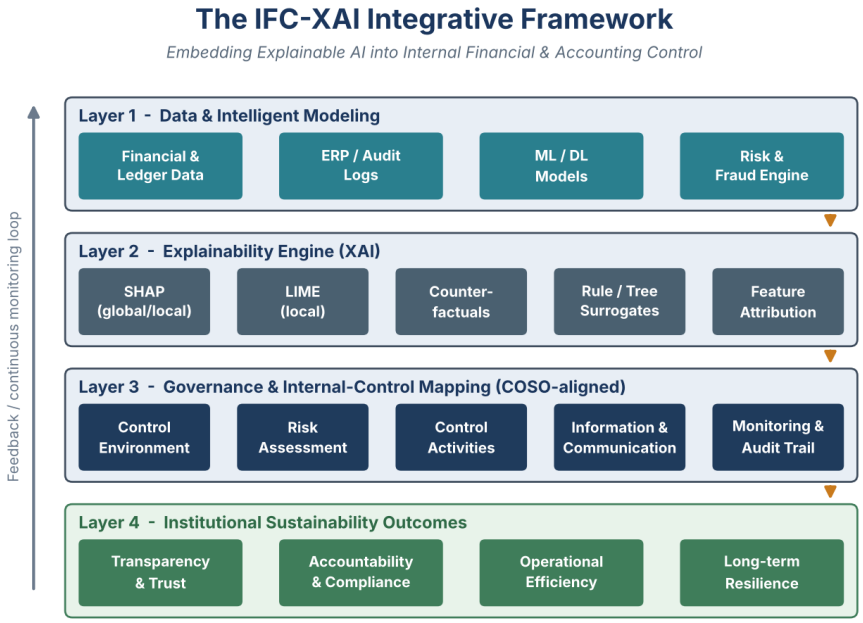


Fig. 1. The IFC-XAI framework for embedding explainable AI in internal financial and accounting control.

- Layer 1, Data and intelligent modeling: financial and ledger data, ERP and audit logs, ML/DL models, and risk and fraud engines.
- Layer 2, Explainability engine: SHAP, LIME, counterfactual explanations, rule- or tree-based surrogates and feature attribution, used together because each covers what the others miss.
- Layer 3, Governance and internal-control mapping: explanation outputs are distributed across the five COSO components. This layer is the main conceptual contribution of the framework.
- Layer 4, Institutional sustainability outcomes: transparency and trust, accountability and compliance, operational efficiency, and long-term resilience.

The feedback loop makes the framework cyclical. What monitoring and the sustainability outcomes reveal flows back into the data and models, so a model is retrained, or its explanation criteria revised, when the risk pattern shifts. Institutional sustainability is treated here as a conceptual outcome of the framework. The empirical part of this study did not measure it.

4.1 Operationalizing Layer 3 through the five COSO components

Table 3. Role of explainability in each COSO component under IFC-XAI.

COSO component	Proposed role of XAI outputs
Control environment	An AI governance committee within the audit-committee structure approves models used in financial control and mandates a documented explanation (e.g., SHAP and LIME outputs) before activation.
Risk assessment	Machine-learning models classify financial risk, and the global SHAP analysis is attached to the annual risk-assessment document.
Control activities	Features with the highest SHAP weight (e.g., checking-account status, loan duration, installment rate) are tied to automated control limits that trigger alerts when exceeded.
Information and communication	A dashboard presents local SHAP explanations in plain language to finance managers and auditors, translating numeric attributions into understandable statements.
Monitoring activities	Periodic monitoring of the stability of feature importance over time; a material change in ranking signals a shift in the risk pattern or the need to recalibrate, and acts as an early-warning indicator.

Read together, these roles turn the distribution of explanatory weight reported in Section 5.4 from an observation into a design choice. Layer 2 outputs feed information and communication as well as monitoring, so explanation becomes part of an ongoing control cycle.

5. Results

5.1 Predictive performance

Table 4 gives test-set performance (n = 300) at the default threshold of 0.50. Logistic regression reached the highest AUC (0.801), ahead of random forest (0.797), gradient boosting (0.765) and the decision tree (0.633). Random forest had the highest precision (0.757) and the lowest recall (0.311), so it was cautious about labeling a case as risky. The single decision tree did worst, which is what overfitting would lead one to expect.

Table 4. Test-set performance of the four classifiers (threshold = 0.50).

Model	Accuracy	Precision	Recall	F1-score	AUC
Logistic Regression	0.7833	0.6761	0.5333	0.5963	0.8011
Random Forest	0.7633	0.7568	0.3111	0.4409	0.7971
Gradient Boosting	0.7633	0.6557	0.4444	0.5298	0.7651
Decision Tree	0.6567	0.4393	0.5222	0.4772	0.6326

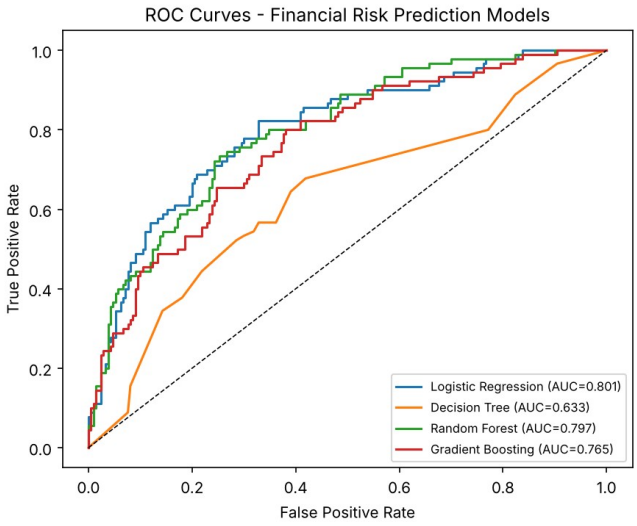


Fig. 2. ROC curves of the four classifiers on the test set.

In the confusion matrix of the logistic regression model (Fig. 3) there are 187 true negatives and 48 true positives. Forty-two risky cases were classed as non-risky (type II errors) and 23 non-risky cases as risky (type I errors). For internal control the type II error usually costs more, because it is a risk that the control failed to catch.

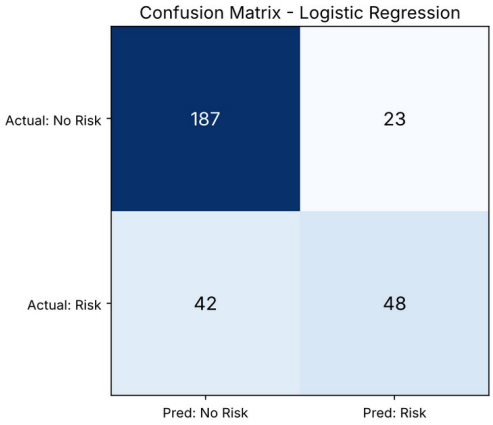


Fig. 3. Confusion matrix of the logistic regression model on the test set.

5.2 Robustness: cross-validation and confidence interval

Table 5 gives mean performance over 15 evaluations (5 folds × 3 repeats). Random forest (AUC = 0.7929) and logistic regression (AUC = 0.7806) came out close, with random forest slightly ahead on average. A paired t-test on fold-level AUC found that the difference (0.0122) was not significant at the 5% level (t = -2.07, p = 0.058), so the lead of logistic regression on the hold-out split should not be read as superiority. The 95% bootstrap confidence interval for its test AUC ran from 0.742 to 0.856. That is wide, as one would expect with so small a test sample.

Table 5. Repeated stratified cross-validation results (5 folds × 3 repeats; mean values).

Model	Accuracy	Recall	F1-score	AUC	AUC SD
Logistic Regression	0.7497	0.4767	0.5334	0.7806	0.0309
Decision Tree	0.6987	0.3822	0.4240	0.6833	0.0407
Random Forest	0.7570	0.2911	0.4173	0.7929	0.0289
Gradient Boosting	0.7517	0.4878	0.5403	0.7750	0.0307

5.3 Decision-threshold analysis

Recall at the default threshold was modest, so the effect of lowering the threshold on the logistic regression model was examined (Table 6). At 0.30, recall climbed from 0.533 to 0.733 and missed risky cases dropped from 42 to 24, but false alarms grew from 23 to 59. The right operating point depends on how an institution weighs a missed risk against the extra review work. Under IFC-XAI this is a management decision to be documented, not a technical default left unexamined.

Table 6. Effect of the decision threshold on the logistic regression model (test set).

Threshold	Accuracy	Precision	Recall	F1	Missed risks (FN)	False alarms (FP)
0.5	0.7833	0.6761	0.5333	0.5963	42	23
0.4	0.7500	0.5773	0.6222	0.5989	34	41
0.3	0.7233	0.5280	0.7333	0.6140	24	59
0.25	0.7133	0.5141	0.8111	0.6293	17	69
0.2	0.6633	0.4663	0.8444	0.6008	14	87

5.4 SHAP attributions and mapping to COSO

Table 7 lists the 15 features with the largest mean absolute SHAP value and the COSO component each was assigned to, and Fig. 4 shows the SHAP summary plot. The most influential features were the absence of a checking account (0.448), low savings (0.327), a critical credit history (0.322), the installment rate (0.317) and loan duration (0.285).

Table 7. Top 15 features by mean |SHAP| and their assigned COSO component.

Rank	Feature	Mean |SHAP|	COSO component
1	No checking account	0.448	Risk Assessment
2	Savings < 100 DM	0.327	Risk Assessment
3	Critical credit history / other credits existing	0.322	Risk Assessment
4	Installment rate	0.317	Control Activities
5	Loan duration (months)	0.285	Control Activities
6	Checking balance < 100 DM	0.228	Risk Assessment
7	Number of existing credits	0.206	Control Activities
8	Personal status: single male	0.199	Control Environment
9	Employment duration 4–7 years	0.183	Control Environment
10	No / unknown property	0.181	Risk Assessment
11	Housing: rent	0.167	Control Environment
12	Loan amount	0.166	Control Activities
13	No / unknown savings account	0.147	Risk Assessment
14	Checking balance 0–200 DM	0.139	Risk Assessment
15	Property: real estate	0.137	Risk Assessment

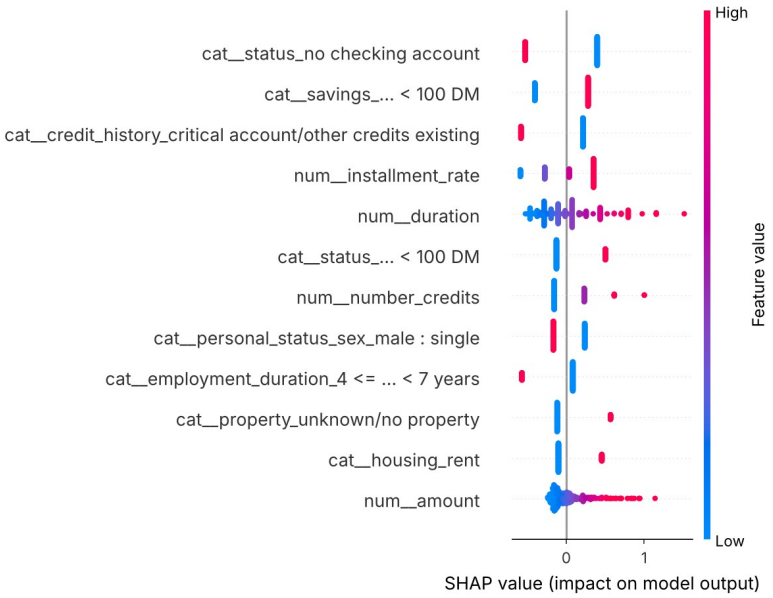


Fig. 4. SHAP summary plot of the most influential features (logistic regression).

Table 8 shows the weight summed by component. Risk Assessment holds more than half (55.9%), followed by Control Activities (28.2%) and Control Environment (15.9%). The model is trained to predict risk, so Risk Assessment leading is hardly surprising. Still, the analysis shows that XAI can work as a diagnostic: it indicates where the model's sensitivity is concentrated and therefore where auditors should look first. Information and communication and monitoring activities received no weight. They

describe how explanations are used and are not input features, and the framework deals with them in its design (Table 3).

Table 8. Distribution of explanatory weight across COSO components.

COSO component	Features	Sum of mean |SHAP|	Share (%)
Risk Assessment	8	1.929	55.9
Control Activities	4	0.974	28.2
Control Environment	3	0.548	15.9

6. Discussion

Munoko et al. (2020) worried that opacity weakens the auditor's independent professional judgment. The results point in the same direction: SHAP produces concrete, quantified evidence that can be examined and filed, which narrows the gap in practice. The statistical tie between the interpretable model and the ensembles also supports the view that chasing maximum accuracy need not cost interpretability, in line with Rudin (2019). That holds only in the sense tested here. The difference between logistic regression and random forest was small and not significant on 1,000 cases, and nothing in these results justifies carrying the conclusion over to other data without further testing.

Two practical points follow. First, an institution starting to invest in XAI could reasonably begin with risk assessment, since that is where most of the explanatory weight sat in this case. Second, the threshold analysis shows that the operating point is a governance matter. A missed risk usually costs more than reviewing a false alarm, so the chosen threshold should be justified, approved and recorded, and XAI outputs can inform that choice.

Fairness deserves a separate remark. The personal-status attribute (single male) is among the 15 most influential features, and many jurisdictions restrict or discourage the use of such attributes in credit decisions. The fact that the explanation brings it to light is a concrete control use of XAI, since sensitive or policy-relevant attributes can be surfaced for review under the control environment. Discrimination was not tested formally, so this should be read as a reason to look closer and not as evidence of bias.

Several recommendations for practice follow from the framework. Audit committees may require an XAI analysis for any model used in control decisions. Explanation tooling should give priority to risk assessment. Dashboards should be built with non-technical users in mind. The stability of feature importance ought to be checked at regular intervals as an early-warning signal. And internal auditors need training in reading SHAP and LIME outputs.

7. Limitations and future work

Several limitations qualify these results.

- Nature and age of the data. German Credit is a public credit dataset of German origin, donated to the UCI repository in 1994, and its amounts are in Deutsche Marks. It does not represent a real internal-control system, the Iraqi market or present-day credit conditions. It is suited to testing the framework and not to pricing loans or predicting default locally.
- Sample size. With 1,000 cases (300 in the test set), confidence intervals are wide and generalization is limited.
- COSO mapping. The assignment of features to components follows the author's functional classification. Auditors or control experts have not validated it, and a different assignment could change the reported shares.

- Recall. At the default threshold recall was 0.533, and it improves only by accepting more false alarms (Table 6).
- Model comparison. The differences between logistic regression and random forest are not statistically significant, and no hyperparameters were tuned.
- Explanation technique. Only SHAP was applied, and it was not compared empirically with LIME or counterfactual explanations.
- Framework validation and sustainability. Practitioners have not tested the framework in the field, and institutional sustainability was not measured.

Future work should apply IFC-XAI to real institutional control data, compare XAI techniques for stability and consistency, validate the COSO mapping with expert panels, assess fairness formally, and measure in the field whether explanatory dashboards improve the quality of auditors' decisions.

8. Conclusion

This paper proposed IFC-XAI, a governance framework that joins explainable AI to the COSO internal-control framework, and tested its feasibility on 1000 credit cases. An inherently interpretable model performed about as well as more complex ones (AUC = 0.801 on the hold-out set and 0.7806 under cross-validation). SHAP gave a quantitative way to relate model behavior to control components, with Risk Assessment holding 55.9% of the explanatory weight. The threshold analysis showed that calibrating a model can be handled as a documented governance decision. The findings are limited by the nature of the data and by the unvalidated mapping, and they should be tested on real institutional data before practical conclusions are drawn.

Declarations

CRediT authorship contribution statement. Ali Abdulwahab Yahya Al-Saffar: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Writing – original draft, Writing – review & editing, Visualization.

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Data and code availability. The German Credit dataset is publicly available from the UCI Machine Learning Repository. All code, derived tables and figures are available at https://github.com/AliAlsaffarMosulUniversity/ifc-xai-coso under the MIT license. An interactive dashboard of the results is available at https://alialsaffarmosuluniversity.github.io/ifc-xai-coso/.

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